

Designation: C 1678-24

Standard Specification for Advanced Silicon Nitride Ceramic Bearing Balls

This standard was issued on February 28, 2024; the version of record is continuously maintained.

1. Scope

1.1 This specification covers high-performance silicon nitride ceramic balls for use in rolling element bearings

1.2 The silicon nitride ceramic bearing balls covered by this specification are intended for use in aerospace bearings, machine tool spindles, and high-speed rotating machinery requiring low density, high stiffness, and resistance to corrosion.

1.3 The values stated in SI units are to be regarded as standard. No other units of measurement are included in this standard.

1.4 This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.

2. Referenced Documents

2.1 ASTM Standards:

C 1326 Test Method for Knoop Indentation Hardness of Advanced Ceramics

C 1327 Test Method for Vickers Indentation Hardness of Advanced Ceramics

C 1624 Test Method for Adhesion Strength and Mechanical Failure Modes of Ceramic Coatings

ISO 3290 Rolling bearings - Balls

3. Terminology

3.1 Definitions:

3.1.1 silicon nitride—a ceramic material with composition Si_3N_4 produced by hot isostatic pressing

3.1.2 grade number—specification of geometric and surface quality per ISO 3290

3.1.3 surface waviness—the periodic deviation of the surface from perfect sphericity

4. Classification

4.1 silicon nitride ceramic bearing balls shall be classified according to diameter and precision grade.

4.1.1 **Grade 3**—Diameter 3.175-25.4 mm, ultra-precision grade

4.1.2 **Grade 5**—Diameter 3.175-50.8 mm, high-precision grade

4.1.3 **Grade 10**—Diameter 3.175-76.2 mm, standard-precision grade

5. Ordering Information

5.1 Orders for material under this specification shall include the following information:

5.1.1 ASTM designation and year of issue

5.1.2 Grade (3, 5, or 10)

5.1.3 Nominal ball diameter (mm)

5.1.4 Quantity

5.1.5 Lot diameter variation tolerance

5.1.6 Surface finish requirements

6. Materials and Manufacture

6.1 Balls shall be manufactured from silicon nitride powder densified by hot isostatic pressing (HIP).

6.2 Sintering aids shall be limited to yttria (Y₂O₃) and/or alumina (Al₂O₃).

6.3 Final density shall be $\geq 99.5\%$ of theoretical density.

7. Chemical Composition

7.1 The material shall conform to the chemical composition requirements specified in Table 1.

Element	Composition, %
Silicon nitride (Si ₃ N ₄)	$\geq 90\%$
Yttrium oxide (Y ₂ O ₃)	2-8%
Aluminum oxide (Al ₂ O ₃)	2-8%
Metallic impurities	$< 0.1\%$

8. Mechanical and Physical Requirements

Property	Requirement	Test Method
Density	3.20-3.28 g/cm ³	Archimedes
Hardness	≥ 1500 HV1	ASTM C 1327
Fracture toughness	≥ 6.0 MPa·m ^{0.5}	Indentation
Flexural strength	≥ 700 MPa	4-point bend

9. Dimensions and Permissible Variations

9.1 Diameter tolerance shall conform to ISO 3290 for the specified grade.

9.2 Sphericity deviation: Grade 3 ≤ 0.13 μm , Grade 5 ≤ 0.25 μm , Grade 10 ≤ 0.50 μm

9.3 Surface roughness Ra: Grade 3 ≤ 0.010 μm , Grade 5 ≤ 0.020 μm , Grade 10 ≤ 0.032 μm

9.4 Lot diameter variation shall not exceed grade tolerance.

10. Workmanship, Finish, and Appearance

10.1 Balls shall be free from cracks, chips, pits, and scratches. Surface finish shall be achieved through lapping and polishing operations.

11. Sampling

11.1 100% inspection of diameter and surface defects. Destructive testing on representative samples from each manufacturing lot.

12. Test Methods

12.1 **Dimensional Measurement**—Measure diameter using laser micrometry or precision mechanical gauging per ISO 3290.

12.2 **Hardness Testing**—Determine Vickers hardness on polished flat ground on balls per ASTM C 1327.

12.3 **Microstructural Examination**—Examine polished and etched cross-sections to verify grain size and phase distribution.

13. Inspection

13.1 Inspection of the material shall be made as agreed upon between the purchaser and the seller as part of the purchase contract.

14. Rejection and Rehearing

14.1 Material that fails to conform to the requirements of this specification may be rejected. Rejection should be reported to the producer or supplier promptly and in writing.

15. Certification

15.1 A manufacturer's or supplier's certification shall be furnished to the purchaser that the material was manufactured, sampled, tested, and inspected in accordance with this specification and has met the requirements.

16. Product Marking

16.1 Balls shall be supplied in labeled containers indicating diameter, grade, quantity, lot number, and manufacturer.

17. Packaging and Package Marking

17.1 Balls shall be packaged in compartmented trays or bulk containers with protective cushioning. Containers shall be sealed to prevent contamination.

18. Keywords

18.1 silicon nitride; ceramic; bearing balls; high-performance; aerospace; precision